

Prithvish Baidya

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India · Remote-first · Comfortable with US (PT), EU (CET), or async timezones

Systems engineer on the Rollups Team at [Conduit](#), operating and upgrading 100+ production rollups. Co-authored [ERC-7518](#). Built the transaction backend powering [American Express's NFT program](#). Won [ETHIndia 2024](#) (2,500+ hackers, 36 hours). Writing code for 14 years, 5 of them shipping production systems: rollup and transaction infrastructure, cryptography, and protocol tooling.

EXPERIENCE

[Conduit](#) | Software Engineer — Rollups Team

Apr 2026 – Present

Remote

- Operate and upgrade a fleet of 100+ production rollups (OP Stack and Arbitrum Orbit) on Kubernetes: GitOps releases staged from canary to testnets to mainnets, with rendered-diff verification before anything reaches a mainnet.
- Own OP Stack client upgrades end to end: rebase Conduit's forks of the node, batcher and challenger onto each upstream release, re-work patches where upstream has moved, and take mandatory hardfork releases fleet-wide on schedule.
- On-call for production incidents: trace chain stalls, sync failures and settlement issues from metrics and logs down to client source, separate cause from symptom under time pressure, and turn each fix into a chart default or an alert so it does not recur.
- Designed and built a stateless Rust service that stands in for a full node as the sync peer for customer-run nodes, speaking Ethereum's execution and consensus p2p protocols from a rolling in-memory window. Took it from design to fleet-wide production and retired the service it replaced.
- Built on-chain security monitoring (Go) that alerts on contract upgrades, ownership changes and large fund movements across L1 and L2, enabled by default for every rollup. Hardened the audit, logging and alerting of an encrypted data-availability service.

[thirdweb](#) | Software Engineer

Jul 2024 – Nov 2025

Remote

- Built [engine-core](#): Rust transaction backend powering thirdweb's API. 100k+ daily transactions across 2,000+ chains. Nonce management is the hard part, and this makes "mint 10,000 NFTs in one call" actually work.
- Powers [Amex Passport](#), issuing digital collectibles to 140M+ cardholders. Zero failed transactions in production.
- Built [twmq](#): Redis-backed job queue with exactly-once delivery, sub-100ms p99 latency. Handles crash recovery and retries for engine-core.
- Designed **Vault**: non-custodial key management using AWS Nitro Enclaves. Keys live in a TEE with cryptographic attestation. Enterprise custody without the custody risk.

[Zoniq](#) | Lead Architect

Jul 2022 – Jun 2024

Remote

- Led team of 4 engineers. Co-authored [ERC-7518](#): Ethereum standard for compliant real-world asset tokenization. Introduces partition-based semi-fungible tokens with dynamic transfer restrictions.
- Built the first compliant RWA tokenization layer on XRPL.
- Architected enterprise platforms: **Heimdall** (web3 operational layer) and **CompliTO** (compliant security token transfers).

[Biconomy](#) | Software Engineer

Nov 2021 – May 2022

Remote

- Rebuilt gasless transaction infrastructure processing 2M+ meta-transactions/month. Enterprise clients included JP Morgan.
- Built the first multichain **ERC-4337** smart contract wallet (hackathon POC → production in 3 months).

Libertas Network | Senior Fullstack Developer

Mar 2021 – Nov 2021

Remote

- Built decentralized censorship-resistant podcasting platform using IPFS, Node.js, and Ethereum. (Company shut down)

PROJECTS

[zkRail](#) | [ETHIndia 2024 Winner](#) | [Demo Video](#)

Dec 2024

Trustless fiat settlement using ZK proofs over banking APIs. Solver network executes real payments; 150% collateral + game-theoretic bonds prevent fraud. Won 1st place overall (2,500+ hackers, \$50k+ prize pool) plus 4 track bounties. Built in 36 hours solo. [Technical writeup](#).

[t2z](#) | 1st Place ECC Track, Zypherpunk 2025 (\$9k total)

2025

Rust library for Zcash shielded transactions. Compiles Halo 2 ZK proving to 3.7MB WASM (no 50MB parameter downloads). Implements ZIP-374 PCZT with external signer support for hardware wallets. Cross-platform via UniFFI.

[Coredrain](#)

2025

Indexer for Hyperliquid's exotic architecture (HyperCore L1 to HyperEVM). Correlates L1 transfers with EVM transaction hashes. Discovered [undocumented dual-hash convention](#) for system transactions by reverse-engineering nanoreth source. Lazy indexing with block interpolation search.

PUBLICATIONS & OPEN SOURCE

[ERC-7518: Dynamic Compliant Interop Security Token](#), Co-author

Review, *Ethereum EIPs*

[engine-core](#): Production Rust transaction infrastructure for blockchain. Primary author.

2024–2025

[Bun, PR #25514](#): Added S3 Requester Pays support to Bun's native S3 client. Zig, TypeScript.

2025

STACK

Languages / Rust, Go, TypeScript, Solidity

Rollups / OP Stack (op-node, op-batcher, op-challenger, op-reth), Arbitrum Nitro, devp2p, libp2p

Infrastructure / Kubernetes, Helm, Argo CD, Prometheus, Grafana, PostgreSQL, Redis, Kafka, AWS Nitro Enclaves, Cloudflare (Workers, D1, R2, Durable Objects, Queues)

Cryptography / ZK (Halo 2, Noir), Zcash/Orchard, WASM

Frontend / React, Next.js, Astro, Tanstack, Three.js

RECOGNITION

[ETHIndia 2024](#): 1st place overall (2,500+ hackers) plus Polygon, Coinbase, CDP, Base bounties

Dec 2024

[Zypherpunk 2025](#): 1st place ECC track (\$5k), plus Tachyon, Raybot, Network School bounties (\$9k total)

2025

KVPY-SA Fellowship: National science fellowship by Indian Institute of Science. Top 1% of 100,000+ applicants selected for research mentorship.

2019

NTSE Scholar: National Talent Search Examination. Nationwide exam selecting top 1,000 students from 1.5M+ for merit scholarship.

2019